

The significance of azo-reduction in the mutagenesis and carcinogenesis of azo dyes.

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Azo dyes are widely used in textile, printing, cosmetic, drug and food-processing industries. They are also used extensively in laboratories as either biological stains or pH indicators. The extent of such use is related to the degree of industrialization. Since intestinal cancer is more common in highly industrialized countries, a possible connection may exist between the increase in the number of cancer cases and the use of azo dyes. Azo dyes can be reduced to aromatic amines by the intestinal microflora. The mutagenicity of a number of azo dyes is reviewed in this paper. They include Trypan Blue, Ponceau 3R, Ponceau 2R, Methyl Red, Methyl Yellow, Methyl Orange, Lithol Red, Orange I, Orange II, 4-Phenylazo-Naphthylamine, Sudan I, Sudan IV, Acid Alizarin Violet N, Fast Garnet GBC, Allura Red, Ponceau SX, Sunset Yellow, Tartrazine, Citrus Red No. 2, Orange B, Yellow AB, Carmoisine, Mercury Orange, Ponceau S, Versatint Blue, Phenylazophenol, Evan's Blue and their degraded aromatic amines. The significance of azo reduction in the mutagenesis and carcinogenesis of azo dyes is discussed.